

LECTURE 6

The CSS Box Model and Display

Every element is a rectangular box — understanding its size and behavior is the core of CSS layout.

CSC336 Web Technologies

In This Lecture

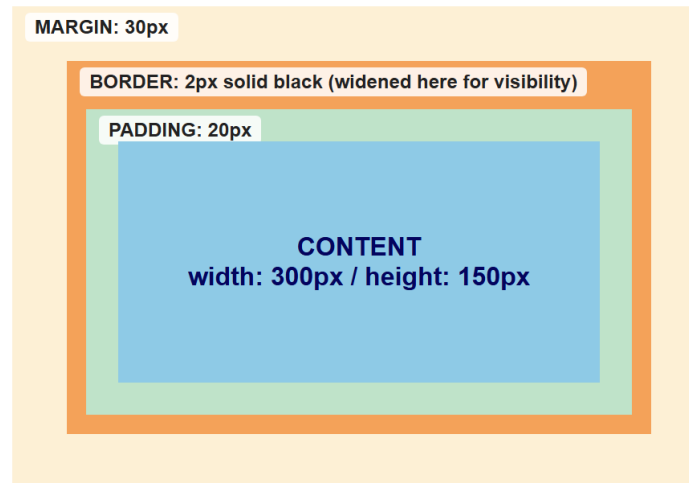
- 1 The four layers of the box model: content, padding, border, and margin
- 2 Margin collapsing, and why it surprises beginners
- 3 box-sizing: content-box vs. border-box
- 4 Controlling size with width, height, and min-/max- constraints
- 5 The display property: block, inline, inline-block, and none
- 6 A real technique: turning a `<ul>` into a horizontal menu with display

Four Nested Layers

- 1 Content — the actual text, image, or other content
- 2 Padding — transparent space inside the border, colored by the background
- 3 Border — a line (or nothing) drawn around the padding
- 4 Margin — transparent space outside the border, never filled with color

The Four Layers, Rendered

```
.content { width: 300px; height: 150px; background: steelblue; }  
.padding { padding: 20px; background: seagreen; }  
.border { border: 20px solid darkorange; }  
.margin { margin: 20px; background: wheat; }
```



Two Adjacent Vertical Margins

```
.box-one { margin-bottom: 30px; }  
.box-two { margin-top: 20px; }
```

You might expect a 50px gap — but margin collapsing means the actual gap is only 30px, the LARGER of the two.

Margin Collapsing, Rendered

First box (margin-bottom: 30px)

Second box (margin-top: 20px)

The visible gap between the two boxes above is 30px — the larger of the two margins — not 50px (30px + 20px), because of margin collapsing.

Only Happens Vertically

WARNING

Margin collapsing applies only to top/bottom margins of block elements in normal flow. It never happens with left/right margins, or with flex, grid, floats, or absolute positioning.

content-box (Default) vs. border-box

```
/* content-box: padding + border ADD to width */  
.a { box-sizing: content-box; width: 300px; padding: 20px; border: 2px solid; }  
/* actual rendered width = 344px */  
  
/* border-box: padding + border are INCLUDED in width */  
.b { box-sizing: border-box; width: 300px; padding: 20px; border: 2px solid; }  
/* actual rendered width = exactly 300px */
```

Same Width, Different Sizing

Same width, padding, and border on both — content-box renders visibly wider than border-box.

box-sizing: content-box

width: 300px + padding: 20px + border: 2px
actual rendered width = 344px

**width: 300px
(content only)**

box-sizing: border-box

width: 300px includes padding + border
actual rendered width = 300px

**width: 300px
(content + padding + border)**

border-box: The Practical Default

TIP

Because border-box makes sizing far more predictable, most real style sheets start with `*`, `*::before`, `*::after` { `box-sizing: border-box;` } as a site-wide reset.

min-/max- Constraints

- max-width caps how wide a box can grow — never wider than 1000px, even on a huge screen
- min-width / min-height set a floor the box will never shrink below
- Together they let a box scale down on narrow screens but stop growing past a sensible limit on wide monitors

block, inline, inline-block, none

- 1 block — starts a new line, fills the full available width (div, p, h1-h6, ul)
- 2 inline — flows within text, width/height are ignored (span, a, strong)
- 3 inline-block — flows inline, but respects width/height/vertical margin (nav links, buttons)
- 4 none — removed from the page entirely, taking up no space at all

All Four Values, Compared

```
.section { display: block; }      /* new line, full width */
.tag     { display: inline; }    /* flows in text, no w/h */
.button  { display: inline-block; /* inline + respects w/h */
          width: 120px; height: 40px; }
.hidden-panel { display: none; } /* removed entirely */
```

display: block

Block 1

Block 2

Block 3

display: inline

Text flows around **Inline A** **Inline B** **Inline C** right in the middle of the sentence, all on one line.

display: inline-block

Button 1

Button 2

Button 3

display: none

Visible before Visible after — notice there is no gap left where the hidden element would have been.

display: none vs. visibility: hidden

```
.gone      { display: none; }      /* removed entirely — no space reserved */  
.invisible { visibility: hidden; } /* invisible, but still takes up its space */
```

Toggling display: none with JavaScript reflows the page (neighbors shift to fill the gap); visibility: hidden leaves a blank gap where the element used to be.

A Row of Equal-Sized Buttons

```
.btn {  
  display: inline-block;  
  width: 120px;  
  text-align: center;  
  padding: 10px 0;  
  margin-right: 8px;  
}
```

inline-block is what lets several fixed-width buttons sit on one line, side by side, without giving up width/height control the way plain inline would.

Turning a `<ul>` Into a Horizontal Menu

```
.menu { list-style: none; margin: 0; padding: 0; }  
.menu li { display: inline-block; margin-right: 20px; }
```

li is block by default — this one change turns a vertical list into a horizontal menu, with no HTML change at all.

Before and After, Rendered

Default: `li` is block

- [Home](#)
- [About](#)
- [Services](#)
- [Contact](#)

```
.menu li { display: inline-block; }
```

Home About Services Contact

Why Build a Menu This Way?

- A nav menu IS, conceptually, a list of links — that's exactly what it is to a screen reader and a search engine
- Screen readers announce it as a list, telling users how many links there are
- The HTML never has to change later — redesign to a dropdown or hamburger menu by changing only CSS

Borders, Radius, and Shadow

```
.card {  
  border: 2px solid #333;  
  border-radius: 8px;  
  box-shadow: 0 4px 8px rgba(0,0,0,0.2);  
}  
  
.avatar { border-radius: 50%; } /* a circle */
```

Consistent Spacing Techniques

- A spacing scale — pick a small set of values (4px, 8px, 16px, 24px) and only ever use those
- CSS custom properties — define `--space-md: 16px` once in `:root`, reuse everywhere with `var(--space-md)`
- Change the whole site's spacing rhythm by editing one value in `:root`

KEY TAKEAWAYS

Lecture 6 in Six Points

- 1 Every box has four layers: content, padding, border, margin — vertically adjacent margins can collapse to the larger value.
- 2 box-sizing: border-box makes padding and border count inside your width, and is the practical default.
- 3 min-/max-width/height add flexible constraints on top of a base size.
- 4 display controls layout behavior: block fills width, inline flows in text, inline-block combines both, none removes the element entirely.
- 5 A `<ul>` styled with `li { display: inline-block; }` is a real way to build a menu without losing list semantics.
- 6 border-radius, box-shadow, and CSS custom properties keep a design finished and consistent.